

Supplementary Information: Time-resolved pairing gap spectroscopy in a quantum simulator of fermionic superfluidity inside an optical cavity

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I. THEORY MODEL

We consider cavity-mediated interactions in an effective three-level system using ^{88}Sr atoms in optical cavity. The cavity axis is in the $\hat{x}$ direction, the quantization axis of atoms is in the $\hat{z}$ direction, and the electric field polarization of the cavity mode is in the $\hat{y}$ direction (see Fig. 1 in the main text). Here we use $|g\rangle$ to denote the $|^1S_0, m=0\rangle$ state of ^{88}Sr atoms, and $|\pm 1\rangle$ for the $|^3P_1, m=\pm 1\rangle$ states. As we discussed below, we can restrict the dynamics in the subspace spanned by $\{|g\rangle, |-1\rangle, |+1\rangle\}$, and consider each atom as an effective three-level system.

The $\hat{y}$ -polarized cavity mode with frequency ω_c is coupled to the atomic transition (frequency ω_a) $|g\rangle \rightarrow (|+1\rangle + |-1\rangle)/\sqrt{2}$, leading to the following atom-light coupling Hamiltonian,

$$\hat{H}_{\text{AL}}/\hbar = g_c \sum_j \left[i\eta_j \hat{a} \frac{|+1\rangle_j \langle g| + |-1\rangle_j \langle g|}{\sqrt{2}} + h.c. \right], \quad (\text{S1})$$

where j is the atomic index, $\hat{a}$ is the annihilation operator of the cavity mode, g_c is the peak atom-cavity coupling strength, and η_j are dimensionless numbers of order 1 that describe the spatial inhomogeneity in the couplings. In a standing wave cavity, we have $\eta_j = \cos(n_j \varphi)$ with $\varphi = \pi \lambda_L / \lambda_c$, where λ_c is the cavity wavelength, λ_L is the lattice wavelength, and n_j is the lattice site index.

Additionally, we consider the Zeeman shift δ_z generated by the external magnetic field,

$$\hat{H}_Z/\hbar = \frac{\delta_z}{2} \sum_j \left[|+1\rangle_j \langle +1| - |-1\rangle_j \langle -1| \right], \quad (\text{S2})$$

as well as the tensor light shift generated by the optical lattice,

$$\hat{H}_T/\hbar = - \sum_j \delta_{T,j} |g\rangle_j \langle g|. \quad (\text{S3})$$

The inhomogeneity in the tensor light shift is originated by the different lattice intensity for different radial modes.

We consider the pump laser has frequency ω_p . In the rotating frame of the pump laser, the full Hamiltonian of this system can be written as

$$\begin{aligned} \hat{H}/\hbar = & g_c \sum_j \left[i\eta_j \hat{a} \frac{|+1\rangle_j \langle g| + |-1\rangle_j \langle g|}{\sqrt{2}} + h.c. \right] + \frac{\delta_z}{2} \sum_j \left[|+1\rangle_j \langle +1| - |-1\rangle_j \langle -1| \right] - \sum_j \epsilon_j |g\rangle_j \langle g| \\ & + \Delta_c \hat{a}^\dagger \hat{a} + \zeta \hat{a}^\dagger + \zeta^* \hat{a}, \end{aligned} \quad (\text{S4})$$

where $\Delta_c = \omega_c - \omega_p$, $\epsilon_j = \omega_a - \omega_p + \delta_{T,j}$, and ζ is the amplitude of the injected field in the cavity.

In this system, it is convenient to choose the atomic basis as bright state $|b\rangle$ (can couple to cavity photon), dark state $|d\rangle$ (cannot couple to cavity photon) and ground state $|g\rangle$ for each atom,

$$|b\rangle = \frac{|1, 1\rangle + |1, -1\rangle}{\sqrt{2}}, \quad |d\rangle = \frac{|1, 1\rangle - |1, -1\rangle}{\sqrt{2}}. \quad (\text{S5})$$

One can define operators $\hat{S}_{\mu\nu,j} = |\mu\rangle_j \langle \nu|$ with $\mu, \nu = \{b, d, g\}$. In terms of these operators, the full Hamiltonian becomes

$$\hat{H}/\hbar = \frac{\delta_z}{2} \sum_j (\hat{S}_{bd,j} + \hat{S}_{db,j}) - \sum_j \epsilon_j \hat{S}_{gg,j} + g_c \sum_j (i\eta_j \hat{S}_{bg,j} \hat{a} - i\eta_j^* \hat{S}_{gb,j} \hat{a}^\dagger) + \Delta_c \hat{a}^\dagger \hat{a} + \zeta \hat{a}^\dagger + \zeta^* \hat{a}. \quad (\text{S6})$$

In addition to the Hamiltonian dynamics, we also consider the dissipation processes such as cavity loss with a rate κ , and spontaneous emission with a rate γ , so the dynamics of this system can be described by the following Lindblad master equation,

$$\frac{d}{dt} \hat{\rho} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}] + \left[\hat{L}_{\text{cav}} \hat{\rho} \hat{L}_{\text{cav}}^\dagger - \frac{1}{2} \{ \hat{L}_{\text{cav}}^\dagger \hat{L}_{\text{cav}}, \hat{\rho} \} \right] + \sum_j \left[\hat{L}_{b,j} \hat{\rho} \hat{L}_{b,j}^\dagger - \frac{1}{2} \{ \hat{L}_{b,j}^\dagger \hat{L}_{b,j}, \hat{\rho} \} \right] + \sum_j \left[\hat{L}_{d,j} \hat{\rho} \hat{L}_{d,j}^\dagger - \frac{1}{2} \{ \hat{L}_{d,j}^\dagger \hat{L}_{d,j}, \hat{\rho} \} \right], \quad (\text{S7})$$

where jump operator for cavity loss is given by

$$\hat{L}_{\text{cav}} = \sqrt{\kappa} \hat{a}, \quad (\text{S8})$$

and the single-particle jump operators for spontaneous emission are given by

$$\hat{L}_{b,j} = \sqrt{\gamma} \hat{S}_{gb,j}, \quad \hat{L}_{d,j} = \sqrt{\gamma} \hat{S}_{gd,j}. \quad (\text{S9})$$

A. Adiabatic elimination of cavity photon

In experiment, Δ_c is the largest frequency scale ($\Delta_c \gg g_c \sqrt{N}$), so we can adiabatically eliminate the cavity photons and obtain an atom-only Hamiltonian. First we expand the annihilation operator of the cavity mode $\hat{a}$ as a sum of coherent state amplitude α (c-number) and quantum fluctuation $\hat{b}$,

$$\hat{a} = \alpha + \hat{b}, \quad \alpha = \frac{-\zeta}{\Delta_c - i\kappa/2}, \quad (\text{S10})$$

so the Hamiltonian becomes

$$\begin{aligned} \hat{H}/\hbar = & \frac{\delta_z}{2} \sum_j (\hat{S}_{bd,j} + \hat{S}_{db,j}) - \sum_j \epsilon_j \hat{S}_{gg,j} + \frac{1}{2} \sum_j i(\Omega \eta_j \hat{S}_{bg,j} - \Omega^* \eta_j^* \hat{S}_{gb,j}) \\ & + g_c \sum_j (i\eta_j \hat{S}_{bg,j} \hat{b} - i\eta_j^* \hat{S}_{gb,j} \hat{b}^\dagger) + \Delta_c \hat{b}^\dagger \hat{b}, \end{aligned} \quad (\text{S11})$$

where $\Omega = -2g_c\zeta/(\Delta_c - i\kappa/2)$. The jump operator for cavity loss becomes

$$\hat{L}_{\text{cav}} = \sqrt{\kappa} \hat{b}. \quad (\text{S12})$$

Then we follow the Reiter-Sørensen approach [1] to obtain the effective Hamiltonian and jump operator,

$$\hat{H}_{\text{eff}} = \frac{\delta_z}{2} \sum_j (\hat{S}_{bd,j} + \hat{S}_{db,j}) - \sum_j \epsilon_j \hat{S}_{gg,j} + \frac{1}{2} \sum_j i(\Omega \eta_j \hat{S}_{bg,j} - \Omega^* \eta_j^* \hat{S}_{gb,j}) - \chi \sum_{jk} \eta_j \eta_k^* \hat{S}_{bg,j} \hat{S}_{gb,k}, \quad (\text{S13})$$

$$\hat{L}_{\text{col}} = \sqrt{\Gamma} \sum_j \eta_j^* \hat{S}_{gb,j}, \quad (\text{S14})$$

where

$$\chi = \frac{g_c^2 \Delta_c}{\Delta_c^2 + \kappa^2/4}, \quad \Gamma = \frac{g_c^2 \kappa}{\Delta_c^2 + \kappa^2/4}. \quad (\text{S15})$$

The effective Hamiltonian contains the exchange interaction between $|g\rangle$ and $|b\rangle$ states, and the effective jump operator generates superradiant decay of the $|b\rangle$ state. The effective master equation of this system is thus given by

$$\frac{d}{dt} \hat{\rho} = -\frac{i}{\hbar} [\hat{H}_{\text{eff}}, \hat{\rho}] + \left[\hat{L}_{\text{col}} \hat{\rho} \hat{L}_{\text{col}}^\dagger - \frac{1}{2} \{ \hat{L}_{\text{col}}^\dagger \hat{L}_{\text{col}}, \hat{\rho} \} \right] + \sum_j \left[\hat{L}_{b,j} \hat{\rho} \hat{L}_{b,j}^\dagger - \frac{1}{2} \{ \hat{L}_{b,j}^\dagger \hat{L}_{b,j}, \hat{\rho} \} \right] + \sum_j \left[\hat{L}_{d,j} \hat{\rho} \hat{L}_{d,j}^\dagger - \frac{1}{2} \{ \hat{L}_{d,j}^\dagger \hat{L}_{d,j}, \hat{\rho} \} \right]. \quad (\text{S16})$$

In this case, it's convenient to calculate the Heisenberg equation of motion for operators $\hat{S}_{\mu\nu,j}$ in large N limit, and take the expectation value to get mean-field equations (we remove the hat of the operators to represent expectation values). Using the commutation relations $[\hat{S}_{\mu\nu,j}, \hat{S}_{\rho\sigma,k}] = (\hat{S}_{\mu\sigma,j}\delta_{\nu\rho} - \hat{S}_{\rho\nu,j}\delta_{\sigma\mu})\delta_{jk}$, we have

$$\begin{aligned}
\frac{d}{dt}S_{gg,j} &= -\frac{1}{2}(\Omega\eta_j S_{bg,j} + \Omega^*\eta_j^* S_{gb,j}) - i\chi \sum_k (\eta_j\eta_k^* S_{gb,k} S_{bg,j} - \eta_j^*\eta_k S_{bg,k} S_{gb,j}) \\
&\quad + \frac{\Gamma}{2} \sum_k (\eta_j\eta_k^* S_{gb,k} S_{bg,j} + \eta_j^*\eta_k S_{bg,k} S_{gb,j}) + \gamma(S_{bb,j} + S_{dd,j}), \\
\frac{d}{dt}S_{bb,j} &= i\frac{\delta_z}{2}(-S_{bd,j} + S_{db,j}) + \frac{1}{2}(\Omega\eta_j S_{bg,j} + \Omega^*\eta_j^* S_{gb,j}) + i\chi \sum_k (\eta_j\eta_k^* S_{gb,k} S_{bg,j} - \eta_j^*\eta_k S_{bg,k} S_{gb,j}) \\
&\quad - \frac{\Gamma}{2} \sum_k (\eta_j\eta_k^* S_{gb,k} S_{bg,j} + \eta_j^*\eta_k S_{bg,k} S_{gb,j}) - \gamma S_{bb,j}, \\
\frac{d}{dt}S_{gb,j} &= -i\frac{\delta_z}{2}S_{gd,j} - \frac{\Omega}{2}\eta_j(S_{bb,j} - S_{gg,j}) - i\chi \sum_k \eta_j\eta_k^* S_{gb,k}(S_{bb,j} - S_{gg,j}) - i\epsilon_j S_{gb,j} \\
&\quad + \frac{\Gamma}{2} \sum_k \eta_j\eta_k^* S_{gb,k}(S_{bb,j} - S_{gg,j}) - \frac{\gamma}{2}S_{gb,j}, \\
\frac{d}{dt}S_{gd,j} &= -i\frac{\delta_z}{2}S_{gb,j} - \frac{\Omega}{2}\eta_j S_{bd,j} - i\chi \sum_k \eta_j\eta_k^* S_{gb,k} S_{bd,j} - i\epsilon_j S_{gd,j} + \frac{\Gamma}{2} \sum_k \eta_j\eta_k^* S_{gb,k} S_{bd,j} - \frac{\gamma}{2}S_{gd,j}, \\
\frac{d}{dt}S_{bd,j} &= i\frac{\delta_z}{2}(S_{dd,j} - S_{bb,j}) + \frac{\Omega^*}{2}\eta_j^* S_{gd,j} - i\chi \sum_k \eta_j^*\eta_k S_{bg,k} S_{gd,j} - \frac{\Gamma}{2} \sum_k \eta_j^*\eta_k S_{bg,k} S_{gd,j} - \gamma S_{bd,j}.
\end{aligned} \tag{S17}$$

Note that the total atom number is conserved in our model, we do not need to include the differential equation for $S_{dd,j}$ because $S_{dd,j} = 1 - S_{gg,j} - S_{bb,j}$. For the numerical simulation, we first turn on Ω to prepare the corresponding initial states, and then turn off Ω and let the system evolve.

B. Mapping to radio-frequency spectroscopy in ultracold fermionic systems

We consider the following fermionic system with three internal states: $|1\rangle$, $|2\rangle$, $|3\rangle$. We assume the s-wave interaction between $|1\rangle$ and $|2\rangle$ is much larger than the others (close to Feshbach resonance), so we can neglect the interactions related to $|3\rangle$ and only consider BCS pairing between $|1\rangle$ and $|2\rangle$ states. We consider a radio-frequency (RF) drive (no momentum transfer) between $|1\rangle$ and $|3\rangle$ states. RF spectroscopy experiment is to scan the frequency of the RF drive, while in our case we fixed the frequency resonating with the bare transition frequency between $|1\rangle$ and $|3\rangle$ states. The Hamiltonian of this system can be written as

$$\hat{H}/\hbar = \sum_{\mathbf{k}} \sum_{\sigma=\{1,2,3\}} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k},\sigma}^\dagger \hat{c}_{\mathbf{k},\sigma} - \chi \sum_{\mathbf{k}\mathbf{k}'} \hat{c}_{\mathbf{k},1}^\dagger \hat{c}_{-\mathbf{k},2}^\dagger \hat{c}_{-\mathbf{k}',2} \hat{c}_{\mathbf{k}',1} + \frac{\delta_z}{2} \sum_{\mathbf{k}} (\hat{c}_{\mathbf{k},3}^\dagger \hat{c}_{\mathbf{k},1} + h.c.). \tag{S18}$$

Similar to the Anderson pseudospin mapping [2], One can define a complete basis for this Hamiltonian as follows,

$$\begin{aligned}
|g\rangle_{\mathbf{k}} &= |n_{\mathbf{k},1} = 0, n_{-\mathbf{k},2} = 0, n_{\mathbf{k},3} = 0\rangle, \\
|b\rangle_{\mathbf{k}} &= |n_{\mathbf{k},1} = 1, n_{-\mathbf{k},2} = 1, n_{\mathbf{k},3} = 0\rangle, \\
|d\rangle_{\mathbf{k}} &= |n_{\mathbf{k},1} = 0, n_{-\mathbf{k},2} = 1, n_{\mathbf{k},3} = 1\rangle.
\end{aligned} \tag{S19}$$

So we can rewrite this Hamiltonian using operators $\hat{S}_{\mu\nu,\mathbf{k}} = |\mu\rangle_{\mathbf{k}}\langle\nu|$ with $\mu, \nu = \{b, d, g\}$, which gives

$$\hat{H}/\hbar = - \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} \hat{S}_{gg,\mathbf{k}} - \chi \sum_{\mathbf{k}\mathbf{k}'} \hat{S}_{bg,\mathbf{k}} \hat{S}_{gb,\mathbf{k}'} + \frac{\delta_z}{2} \sum_{\mathbf{k}} (\hat{S}_{db,\mathbf{k}} + h.c.), \tag{S20}$$

which agrees with the effective Hamiltonian of our three-level system in cavity.

II. ANALYTIC DISCUSSIONS

A. Mean-field derivation of Δ_{SG} in the ideal case

For simplicity, we consider analytic discussions in the case of homogeneous atom-light couplings, $\eta_j = 1$. We further assume the interaction strength χN is much larger than the frequency disorder ϵ_j , so we can drop the ϵ_j terms and restrict the dynamics in the fully symmetric manifold. We then rewrite the Hamiltonian in terms of Schwinger bosons for the three internal states, i.e. $(\hat{a}_g, \hat{a}_b, \hat{a}_d)$,

$$\hat{H} = \frac{\delta_z}{2} (\hat{a}_b^\dagger \hat{a}_d + \hat{a}_d^\dagger \hat{a}_b) - \chi \hat{a}_b^\dagger \hat{a}_g \hat{a}_g^\dagger \hat{a}_b. \quad (\text{S21})$$

Here we set $\Omega = 0$ during the time evolution. In terms of Schwinger bosons, one can easily calculate the energy shift in RF spectroscopy, which is equivalent to the energy difference between $|b\rangle$ and $|d\rangle$ states. Define $\hat{N}_\mu = \hat{a}_\mu^\dagger \hat{a}_\mu$ with $\mu = \{b, d, g\}$, we have $\chi \hat{a}_b^\dagger \hat{a}_g \hat{a}_g^\dagger \hat{a}_b = \chi \hat{N}_b (\hat{N}_g + 1)$. In the large- N limit, we have $E_b - E_d \approx \chi N_g$.

The mean-field approximation for Eq. (S21) is to derive the Heisenberg equation of motion for $(\hat{a}_g, \hat{a}_b, \hat{a}_d)$, and then replace them by c -numbers: $(\hat{a}_g, \hat{a}_b, \hat{a}_d) \sim \sqrt{N}(v_g, v_b, v_d)^T$, where $|v_g|^2 + |v_b|^2 + |v_d|^2 = 1$. So the mean-field equations in terms of v_g, v_b, v_d takes the following form,

$$\begin{aligned} \dot{v}_g &= i\chi N(v_b^* v_b) v_0, \\ \dot{v}_b &= -i \left(\frac{\delta_z}{2} v_d - \chi N(v_g^* v_b) v_b \right), \\ \dot{v}_d &= -i \frac{\delta_z}{2} v_b. \end{aligned} \quad (\text{S22})$$

The first equation implies that $v_g^* v_g = N_g/N = \text{const.}$ Therefore, we can simplify the equations for v_b and v_d as follows,

$$\begin{aligned} \dot{v}_b &= -i \left(\frac{\delta_z}{2} v_d - \chi N_g v_b \right), \\ \dot{v}_d &= -i \frac{\delta_z}{2} v_b, \end{aligned} \quad (\text{S23})$$

which are linear and can be solved analytically. So the general solution is given by

$$\begin{aligned} v_g(t) &= v_g(0) \exp \left[i\chi N \int_0^t v_b^*(t') v_b(t') dt' \right], \\ v_b(t) &= \left[v_b(0) \cos \left(\frac{\omega t}{2} \right) + i v_b(0) \frac{\chi N_g}{\omega} \sin \left(\frac{\omega t}{2} \right) - i v_d(0) \frac{\delta_z}{\omega} \sin \left(\frac{\omega t}{2} \right) \right] \exp \left(i \frac{\chi N_g}{2} t \right), \\ v_d(t) &= \left[v_d(0) \cos \left(\frac{\omega t}{2} \right) - i v_d(0) \frac{\chi N_g}{\omega} \sin \left(\frac{\omega t}{2} \right) - i v_b(0) \frac{\delta_z}{\omega} \sin \left(\frac{\omega t}{2} \right) \right] \exp \left(i \frac{\chi N_g}{2} t \right), \end{aligned} \quad (\text{S24})$$

where

$$\omega = \sqrt{(\chi N_g)^2 + \delta_z^2}. \quad (\text{S25})$$

The expressions for $v_b(t)$ and $v_d(t)$ describe dynamics that formally resemble Rabi oscillation between $|b\rangle$ and $|d\rangle$ with a Rabi frequency δ_z and a (collective) detuning χN_g . This detuning represents the shift observed in rf spectroscopy experiments, so we identify the spectral gap as $\Delta_{\text{SG}} = \chi N_g$.

B. Comparing Δ_{BCS} and Δ_{SG}

The pairing gap Δ_{BCS} is a defined quantity in the BCS model. Intuition can be gained by considering the mean-field ground state, which under an Anderson pseudospin mapping is described by

$$\langle \hat{J}_k^- \rangle = \frac{1}{2} \frac{\Delta_{\text{BCS}}}{\sqrt{(\epsilon_k - \mu)^2 + |\Delta_{\text{BCS}}|^2}}, \quad \langle \hat{J}_k^z \rangle = -\frac{1}{2} \frac{\epsilon_k - \mu}{\sqrt{(\epsilon_k - \mu)^2 + |\Delta_{\text{BCS}}|^2}}, \quad (\text{S26})$$

with self-consistent equations

$$\Delta_{\text{BCS}} = \chi \sum_k \langle \hat{J}_k^- \rangle = \frac{1}{2} \sum_k \frac{\chi \Delta_{\text{BCS}}}{\sqrt{(\epsilon_k - \mu)^2 + |\Delta_{\text{BCS}}|^2}}, \quad (\text{S27})$$

$$\langle \hat{J}^z \rangle = \sum_k \langle \hat{J}_k^z \rangle = -\frac{1}{2} \sum_k \frac{\epsilon_k - \mu}{\sqrt{(\epsilon_k - \mu)^2 + |\Delta_{\text{BCS}}|^2}}. \quad (\text{S28})$$

In this work, we are mainly concerned with the regime $\epsilon_k \rightarrow 0$. In this limit, one can obtain

$$\langle \hat{J}_k^- \rangle = \frac{1}{2} \frac{\Delta_{\text{BCS}}}{\sqrt{\mu^2 + |\Delta_{\text{BCS}}|^2}}, \quad \langle \hat{J}_k^z \rangle = \frac{1}{2} \frac{\mu}{\sqrt{\mu^2 + |\Delta_{\text{BCS}}|^2}}, \quad (\text{S29})$$

and the self-consistent equations become

$$\frac{\chi N}{2} = \sqrt{\mu^2 + |\Delta_{\text{BCS}}|^2}, \quad \mu = \chi \langle \hat{J}^z \rangle. \quad (\text{S30})$$

Therefore, we can express the pairing gap Δ_{BCS} magnitude by

$$|\Delta_{\text{BCS}}| = \chi \sqrt{\left(\frac{N}{2}\right)^2 - \langle \hat{J}^z \rangle^2}. \quad (\text{S31})$$

In comparison, the spectral gap Δ_{SG} can be expressed in the limit where $\epsilon_k \rightarrow 0$ by

$$\Delta_{\text{SG}} = \chi \langle \hat{N}_g \rangle = \chi \left(\frac{N}{2} - \langle \hat{J}^z \rangle \right) = \sqrt{|\Delta_{\text{BCS}}|^2 + \mu^2} - \mu. \quad (\text{S32})$$

Beyond the limit $\epsilon_k \rightarrow 0$, the RF spectrum based on ideal BCS model shows an RF absorption threshold at $\omega = \chi \langle \hat{N}_g \rangle$ (e.g., see Eq. (2.31) in R. Haussmann *et al.*, *Phys. Rev. A* **80**, 063612 (2009) and “Ultracold Fermi Gases”, Proc. of the Varenna ‘Enrico Fermi’ Summer School 2006, W. Ketterle, M. Inguscio and C. Salomon). The size of the mean-field collective detuning between $|b\rangle$ and $|d\rangle$ will therefore be smaller than $\chi \langle \hat{N}_g \rangle$ outside of the fully collective regime. Likewise, when $J < N/2$, $\Delta_{\text{BCS}} = \chi \langle \hat{J}^- \rangle$ becomes smaller and is roughly equal to $\chi \sqrt{J^2 - \langle \hat{J}^z \rangle^2}$.

Our initial preparation scheme with tunable Rabi pulse area can be interpreted as preparing the mean-field ground state in the limit $\epsilon_k \rightarrow 0$ with fixed $\langle \hat{J}^z \rangle$ (fixed number of Cooper pairs). It follows that we can measure $|\Delta_{\text{BCS}}|$ of this ground state looking at the measured value of $|\Delta_{\text{BCS}}(t)|$ in Fig. 2(b) of the main text at short times, immediately after initialization. In contrast, Δ_{SG} represents the $\delta_z \rightarrow 0$ limit of the oscillation frequency of the curves in Fig. 2(b), as shown in the right panel of Fig. 2(c).

III. EXPERIMENTAL DETAILS

A. The QND Hamiltonian

Eq. (S4) provides a full description of relevant features of the system. However, the effective Hamiltonian derived in Eq. (S13) does not include the additional atom-cavity Hamiltonian term introduced in the main text, which describes the key physics behind our nondestructive cavity probe. Where does the additional term come from?

The effective Hamiltonian description in Sec. I A relies on an adiabatic elimination step that assumes all relevant dynamics occurs close to DC in the rotating frame of the pump laser, which is assumed to be resonant with the atomic transition. In this picture, because the cavity resonance frequency is sufficiently far-detuned, we conclude that the cavity field responds to the atoms “driving” the mode very quickly and thus can adiabatically eliminate the field. However, in the presence of an applied cavity probe laser close to resonance with the cavity (and not the atoms), the system can respond in two frequency bands, complicating this picture.

We describe the pump field ζ using two terms: $\zeta = \zeta^{(a)} + \zeta^{(c)}e^{-i\Delta_c t}$, near resonance with the atomic transition and cavity mode respectively. The two fields can vary in time, but we can cleanly separate the responses if we assume that $|\dot{\zeta}^{(\sigma)}| \ll \Delta_c |\zeta^{(\sigma)}|$, such that the Fourier transform of the field exhibits two well-separated regions centered at DC and Δ_c . In experimental terms, $\zeta^{(a)}$ represents the initial laser drive that excites the atoms. $\zeta^{(c)}$ describes the cavity probe, which is described in detail in the next section. We can characterize the mean-field response of the system to these two separate drives by setting up the following ansatz:

$$\begin{aligned}\langle \hat{a} \rangle &= a := a^{(a)} + a^{(c)}e^{-i\Delta_c t} \\ \langle \hat{S}_{gb,j} \rangle &= S_{gb,j} := S_{gb,j}^{(a)} + S_{gb,j}^{(c)}e^{-i\Delta_c t} \\ \langle \hat{S}_{gd,j} \rangle &= S_{gd,j} := S_{gd,j}^{(a)} + S_{gd,j}^{(c)}e^{-i\Delta_c t}\end{aligned}\tag{S33}$$

The other atomic observables ($\hat{S}_{bd,j}$, $\hat{S}_{bb,j}$, $\hat{S}_{gg,j}$, and $\hat{S}_{dd,j}$) do not represent optical coherences and thus do not exhibit a response in the two (optical frequency) bands, which we will call the “atomic band” and the “cavity band.” **Fig. S1(a) demonstrates that we see well-separated atomic and cavity band responses on our heterodyne detector, justifying our treatment.** In principle, the responses to the two bands can be coupled because the atomic degrees of freedom are highly nonlinear. However, if we assume any dynamics from the cavity band are sufficiently weak and that Δ_c represents the largest dynamical frequency scale in the system, to leading order the responses are decoupled. Experimentally, this represents the limit taken by quantum nondemolition (QND) probes, which are designed to extract information without significantly perturbing the atoms.

In the QND limit, the atomic band responses are well-described by Sec. I A. We can calculate the mean-field cavity band responses by applying the Heisenberg equation of motion to Eq. (S4) and separating out the frequency response at Δ_c . This yields the following set of equations:

$$\begin{aligned}\frac{d}{dt}a^{(c)} &= -\frac{\kappa}{2}a^{(c)} - g_c \sum_j \eta_j S_{gb,j}^{(c)} - i\zeta^{(c)} \\ \frac{d}{dt}S_{gb,j}^{(c)} &= [i(\Delta_c - \epsilon_j) - \frac{\gamma}{2}]S_{gb,j}^{(c)} - i\frac{\delta_z}{2}S_{gd,j}^{(c)} - g_c\eta_j a^{(c)}(S_{bb,j} - S_{gg,j}) \\ \frac{d}{dt}S_{gd,j}^{(c)} &= [i(\Delta_c - \epsilon_j) - \frac{\gamma}{2}]S_{gd,j}^{(c)} - i\frac{\delta_z}{2}S_{gb,j}^{(c)} - g_c\eta_j a^{(c)}S_{bd,j},\end{aligned}\tag{S34}$$

Note that, unlike with the atomic band, equations in the cavity band can have a macroscopic cavity mode population, and instead the atomic coherences adiabatically follow the cavity since they are off resonance by roughly Δ_c . Therefore, we can set the time derivatives of the atomic coherences to 0. Solving the coupled equations to leading order in Δ_c^{-1} yields:

$$\begin{aligned}S_{gb,j}^{(c)} &= \frac{g_c\eta_j a^{(c)}}{[i(\Delta_c - \epsilon_j) - \frac{\gamma}{2}]^2 + \frac{\delta_z^2}{4}} \left(i\frac{\delta_z}{2}S_{bd,j} + [i(\Delta_c - \epsilon_j) - \frac{\gamma}{2}](S_{bb,j} - S_{gg,j}) \right) \\ &= -i\frac{g_c}{\Delta_c}\eta_j a^{(c)}(S_{bb,j} - S_{gg,j}) + \mathcal{O}(\Delta_c^{-2}) \\ S_{gd,j}^{(c)} &= \frac{1}{[i(\Delta_c - \epsilon_j) - \frac{\gamma}{2}]} \left(i\frac{\delta_z}{2}S_{gb,j}^{(c)} + g_c\eta_j a^{(c)}S_{bd,j} \right) \\ &= -i\frac{g_c}{\Delta_c}\eta_j a^{(c)}S_{bd,j} + \mathcal{O}(\Delta_c^{-2}).\end{aligned}\tag{S35}$$

We can then eliminate the atomic observables from the equation of motion for the classical field $a^{(c)}$ to obtain:

$$\begin{aligned} \frac{d}{dt}a^{(c)} &= -\frac{\kappa}{2}a^{(c)} + i\frac{g_c^2}{\Delta_c} \sum_j \eta_j^2 (S_{bb,j} - S_{gg,j})a^{(c)} - i\zeta^{(c)} \\ &= [2i\chi_0 \sum_j \eta_j^2 J_j^z - \frac{\kappa}{2}]a^{(c)} - i\zeta^{(c)}, \end{aligned} \quad (\text{S36})$$

where $\chi_0 \approx g_c^2/\Delta_c$ is the spin-exchange interaction strength (for peak couplers) taken to the limit $\Delta_c \gg \kappa/2$, and $J_j^z = \frac{1}{2}(S_{bb,j} - S_{gg,j})$ is the inversion for atom j along the bright-ground transition. We see that, at least at the mean-field level, the presence of atoms in the cavity acts as an effective frequency shift. Experimental intuition comes from recognizing that the near-resonant atoms act as a gas with an inversion-dependent index of refraction, which changes the cavity resonance frequency.

Using adiabatic elimination techniques (for instance, by applying the Reiter-Sørensen approach [1] for each m_J subspace separately), it is possible to promote this mean-field intuition into an effective QND Hamiltonian for the cavity field:

$$\hat{H}_{\text{QND}} = \left(\Delta_c - 2\chi_0 \sum_j \eta_j^2 \hat{J}_j^z \right) \hat{a}^\dagger \hat{a}. \quad (\text{S37})$$

We can follow definitions from the Supplemental Material of a previous work [3] and define an effective “weighted” inversion operator of the form

$$\hat{J}^{z'} := \frac{\sum_j \eta_j^2 \hat{J}_j^z}{\frac{1}{N} \sum_j \eta_j^2}, \quad (\text{S38})$$

defined such that $\hat{J}^{z'}$ takes values between $\pm N/2$. Combined with an rms interaction strength $\chi' = \chi_0(\frac{1}{N} \sum_j \eta_j^2)$, Eq. (S37) simplifies to $\hat{H}_{\text{QND}} = (\Delta_c - 2\chi' \hat{J}^{z'}) \hat{a}^\dagger \hat{a}$, which is the form presented in the main text. In the experiment, the atoms exhibit couplings characterized by $\eta_j = \cos(\varphi_j)$ with uniformly distributed phases $\varphi_j \in [0, 2\pi)$. In this case, the weighted definitions evaluate to $\hat{J}^{z'} = 2 \sum_j \eta_j^2 \hat{J}_j^z$, $\chi' = \chi_0/2$. Note that both χ and $\hat{J}^z$ in the main text always refer to the primed quantities defined here, which reflects weighting by cavity coupling.

B. The cavity probe

In order to measure the cavity resonance frequency $\omega_c(t)$, we apply a series of probe pulses at the cavity by sending RF pulses into a fiber phase modulator to periodically create an FM sideband nominally resonant with the cavity. To shape each RF pulse, we program an arbitrary waveform generator to produce an RF pulse of the form

$$V(t) = V_{\text{env}}(t) \cos(\tilde{\omega}t); \quad V_{\text{env}}(t) = V_0 e^{-\frac{t^2}{2\tau^2}}; \quad \tau = 50 \text{ ns}, \quad (\text{S39})$$

where $\tilde{\omega} = \omega_{c0} - \omega_l$ is the RF frequency necessary to generate a first-order sideband at the unshifted cavity resonance ω_{c0} , given input light with frequency ω_l . In our case, $\tilde{\omega}/2\pi = -60$ MHz, which is large enough that $V_{\text{env}}(t)$ varies slowly compared to $\tilde{\omega}$ ($|\tilde{\omega}|\tau \gg 1$). The induced FM modulation therefore results in resolved sidebands of light, which follow a Jacobi-Anger expansion:

$$\begin{aligned} E(t) &= E_0 e^{i\omega_l t} \exp\left(i \frac{V_{\text{env}}(t)}{V_\pi} \pi \cos(\tilde{\omega}t)\right) \\ &= E_0 \sum_{n=-\infty}^{\infty} i^n J_n\left(\frac{V_{\text{env}}(t)}{V_\pi} \pi\right) e^{i(\omega_l + n\tilde{\omega})t}, \end{aligned} \quad (\text{S40})$$

where J_n is the n^{th} -order cylindrical Bessel function, and V_π is the half-wave voltage of the fiber phase modulator. In our case, $V_0/V_\pi \approx 0.23$, meaning the modulator is driven close to the linear excitation regime (i.e., $J_1(\frac{V_{\text{env}}(t)}{V_\pi} \pi) \approx \frac{V_{\text{env}}(t)}{V_\pi} \frac{\pi}{2}$). As a result, we assume that the electric field of the first-order FM sideband roughly inherits the shape described by $V_{\text{env}}(t)$.

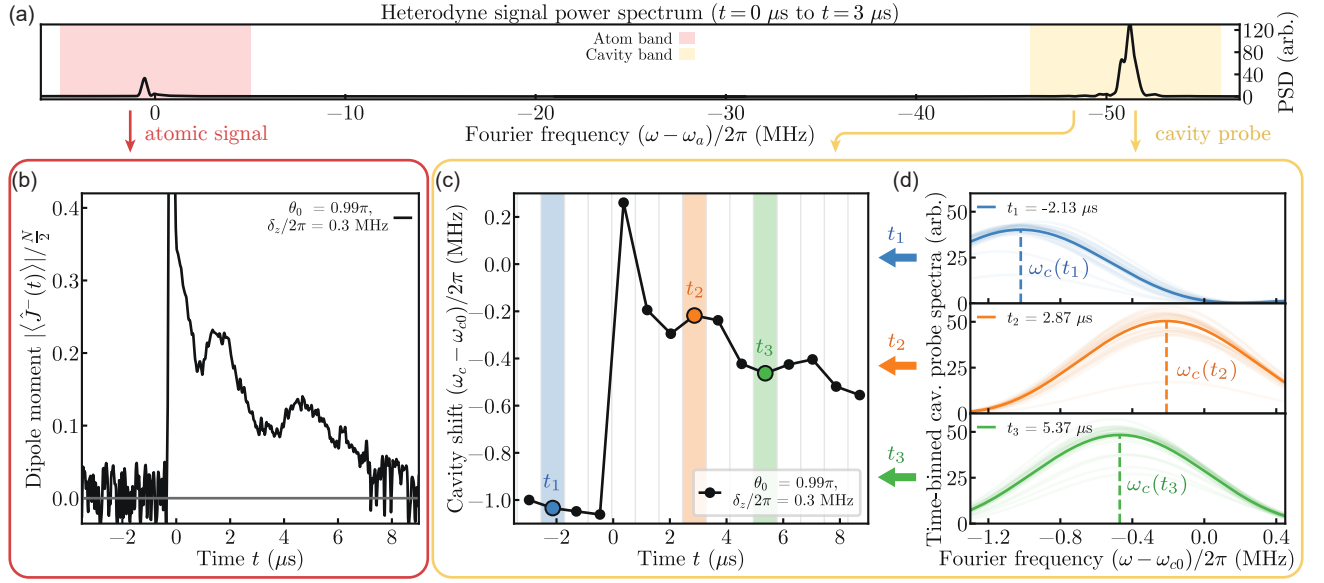


FIG. S1. Simultaneously measuring $\langle \hat{J}^- \rangle$ and $\langle \hat{J}^z \rangle$. (a) Power spectrum of the full signal measured in our heterodyne detector in a time bin from $t = 0 \mu\text{s}$ to $t = 3 \mu\text{s}$. The atomic and cavity bands (shaded regions) are well-separated in Fourier space. (b) From the atomic band, we measure the collective dipole moment $\langle \hat{J}^- \rangle$ vs. time after the initialization drive pulse. Shown here is a trace with Rabi angle $\theta_0 = 0.99\pi$ and Zeeman splitting $\delta_z/2\pi = 0.3 \text{ MHz}$. The inferred $\langle \hat{J}^- \rangle$ becomes large in the roughly 200 ns during the initialization pulse, right before $t = 0$. We attribute this to detection of the initialization light on our detector and therefore disregard our measurements at that time. (c) From the cavity band, we measure the cavity frequency ω_c over time (black points, with three example points highlighted in color). Rescaling by a factor of 2χ allows us to infer the effective atomic inversion $\langle \hat{J}^z(t) \rangle = (\omega_c(t) - \omega_{c0})/2\chi$ along the $|b\rangle$ to $|g\rangle$ transition. We accomplish this measurement by dividing the raw heterodyne signal in this band into time bins (here, $t_{\text{bin}} = 867 \text{ ns}$ for clarity, although in the main text we use $t_{\text{bin}} = 67 \text{ ns}$ for better time resolution). Three example bins are shaded to illustrate the process. (d) For each time bin, we calculate the power spectrum of the heterodyne signal. Here, we plot an averaged response (solid curves) over many instances of the experiment (faint curves) to extract a single estimate of the center frequency, $\omega_c(t_{\text{bin}})$ (dashed line). Our full analysis involves bootstrap resampling and aggregation over frequency estimates of individual runs to obtain a statistical uncertainty (not displayed).

The Fourier spectrum of V_{env} is also a Gaussian, with a standard deviation of $\sigma = 1/\tau \approx 2\pi \times 3.18 \text{ MHz}$ and a HWHM of $\sigma\sqrt{2\ln 2} \approx 2\pi \times 3.75 \text{ MHz}$. This means that if $\omega_c(t) - \omega_0 = 2\pi \times \pm 3.75 \text{ MHz}$ at the time of the pulse, the injected electric field will be half as big as it would have been if the cavity were unshifted, resulting in a 50% reduction in signal-to-noise. In other words, the pulse ensures that the probe will have at least 50% of the maximum signal-to-noise over a 7.5 MHz band centered at ω_{c0} . Over time, the light inside the cavity will leak out, with a $1/e$ time constant of $2/\kappa \approx 2.1 \mu\text{s}$ for the electric field. During this time, the cavity field will evolve according to Eq. (S36) with $\zeta^{(c)} = 0$, as long as changes in $\hat{J}^{z'}$ occur much more slowly than Δ_c , the scale separating the atomic and cavity bands. Therefore, as the light leaks out of the cavity, its frequency is modulated to match the dynamics of $\hat{J}^{z'}$. In other words, the frequency of the cavity photons adiabatically follows the cavity resonance frequency $\omega_c(t)$, since there are no other nearby eigenmodes to mix with. This is illustrated in Fig. S1(a): the cavity response (in the yellow cavity band) is well-split from the atomic transition frequency (in the red atomic band). Other cavity resonances such as different longitudinal modes are split by even larger detunings ($\omega_{\text{FSR}}/2\pi = 3.715 \text{ GHz}$).

We repeat the pulse described above every $5 \mu\text{s}$ in order to keep the cavity populated with photons. In between pulses, we detect the cavity field in transmission by beating it against a local oscillator and measuring in heterodyne to infer a complex electric field amplitude as a function of time. We then time bin the trace into chunks of length $t_{\text{bin}} = 67 \text{ ns}$ (15 bins per μs , compared with a 60 MS/s acquisition rate for the scope), similar to the binning shown in Fig. S1(c) which uses $t_{\text{bin}} = 867 \text{ ns}$ for clarity. Within each time bin, we estimate the ω_c by calculating the peak Fourier response, as shown in Fig. S1(d). These fitted frequencies are then used to reconstruct the time-dependent behavior of ω_c (Fig. S1(c)).

The precision of this probe relies heavily on signal-to-noise considerations. We operate at a power where the maximum intracavity photon number is roughly $M_c \approx 30 \times 10^3$. Due to imperfect quantum efficiency, we effectively collect about 40 photons per time bin per shot of the experiment, which does not allow for a high precision in frequency

estimation. To improve this, we repeat the experiment 8 times within a single loading sequence of the experiment, over a time period of roughly 2 ms. In postprocessing, we average the complex electric field phasors from each of these experimental shots and then estimate the frequency, which multiplies the effective photon number by 8. Then, we repeat this loading sequence for a total number of times $n_{\text{seq}} = 100$. Since we cannot ensure phase stability between loading sequences ($t_{\text{seq}} = 3$ s), we could not directly average the complex phasors. Instead, we process each loading sequence separately to obtain n_{seq} estimates of the cavity resonance frequency per time bin, which we then average down to increase precision by a factor of $\sqrt{n_{\text{seq}}} = 10$. We empirically find that the spread in frequency estimates is around 200 kHz at maximum intracavity power, giving us an estimated maximum frequency precision of $\Delta\omega_c/2\pi = 20$ kHz. The uncertainty region presented in the main text comes not from this estimated precision but instead from performing a bootstrap resampling on the n_{seq} frequency estimates per time bin ($n_{\text{boot}} = 100$) and calculating the standard deviation over resampled estimates. Finally, since the signal-to-noise decreases exponentially between pulses, we repeat the experiment with the cavity probe pulses applied with a $2.5 \mu\text{s}$ offset. In post-processing, we stitch together the two experiments to optimize the signal size in each time bin. We do not observe any distortion of the dynamics by applying the cavity probe pulses at different times.

In principle, we could improve the signal-to-noise of the probe by sending in more photons; however, if the probe becomes too strong it will start to affect the system dynamics (which we do not want). Eq. (S37) can be interpreted as a shift of the atomic transition frequency by a characteristic amount $\chi'\langle\hat{a}^\dagger\hat{a}\rangle = \chi'M_c$, where M_c is the number of intracavity photons. We want this to be small compared with the frequencies of the Hamiltonian of interest, such as $\chi'N \approx 2\pi \times 1$ MHz. For the estimated intracavity photon number M_c , we calculate a maximum frequency scale of $\chi'M_c \approx (2\pi \times 35 \text{ kHz}) \ll \chi'N$. The signal-to-noise could be further improved by increasing the quantum efficiency of our system, which is estimated to be around 0.02. Finally, although we don't believe it to be true in our system, laser frequency noise in the local oscillator could be a limiting factor if it is larger than the shot-noise-limited precision of the probe. Using a laser with narrower linewidth could then also benefit the probe.

C. Effects of inhomogeneous coupling on the spectral gap

As outlined in the main text, for a system with homogeneous atom-light coupling, an initial state in the maximally symmetric manifold (i.e., $J = \frac{N_b + N_g}{2}$) experiences a spectral gap equal to $\Delta_{\text{SG}} = \chi N_g$. The presence of inhomogeneous coupling (see Sec. I) disrupts this picture. It is hard to analytically calculate the spectral gap Δ_{SG} in the case of inhomogeneous coupling. Instead, we rely on numerically simulating Δ_{SG} with the same analysis used on the experimental data. Additionally, we compare Δ_{SG} against what we believe is a reasonable analog for χN_g for an inhomogeneously coupled system. This analog replaces N_g with an estimate of the ground state population weighted by cavity coupling, which we will call N'_g (or, in an abuse of notation, just N_g in the main text):

$$N'_g := \frac{\sum_j \eta_j^2 N_{g,j}}{\frac{1}{N} \sum_j \eta_j^2}, \quad (\text{S41})$$

normalized such that when all atoms are in the ground state, $N'_g = N$. The intuition for why this seems like a good choice comes from Sec. III A, where the weighted atomic inversion $\hat{J}^{z'}$ turns out to be the correct observable to determine shifts in the cavity resonance frequency.

However, this quantity carries an additional complication: while the unweighted ground state atom number N_g is preserved under the spin-exchange interaction (the number of excitations does not change), N'_g is not. This is because when atoms with different cavity coupling strengths exchange excitations, how the excitation is weighted also changes. This can be seen in Fig. S2a, which shows numerical simulations of N'_g for several different drive angles θ_0 (defined as the angle experienced by atoms with maximum coupling). We see that N'_g oscillates for a few μs after initialization, sometimes drastically changing from the $t = 0$ value. Interestingly, in simulations that neglect dissipation (dashed curves), N'_g appears to quickly approach a steady state, where the system attains a sort of detailed balance in the exchange between classes of atoms with different coupling strengths. With dissipation (solid curves), N'_g decays to N at long times.

Fig. S2b shows how various measurements of N'_g differ as a function of drive angle θ_0 . For homogeneous coupling (gray dashed line), $N'_g = N \cos(\theta_0/2)^2$ is constant in the absence of dissipation. In the case of inhomogeneous coupling, we assume the atoms uniformly sample a coupling phase φ , resulting in a coupling strength $\eta(\varphi) = \cos \varphi$. In this case, the weighted ground state population at $t = 0$ ($N'_g{}^{\text{init}}$, blue dashed line) is given by:

$$N'_g{}^{\text{init}}(\theta_0) = N \left(\frac{1}{2} + \frac{J_1(\theta_0)}{\theta_0} - J_2(\theta_0) \right). \quad (\text{S42})$$

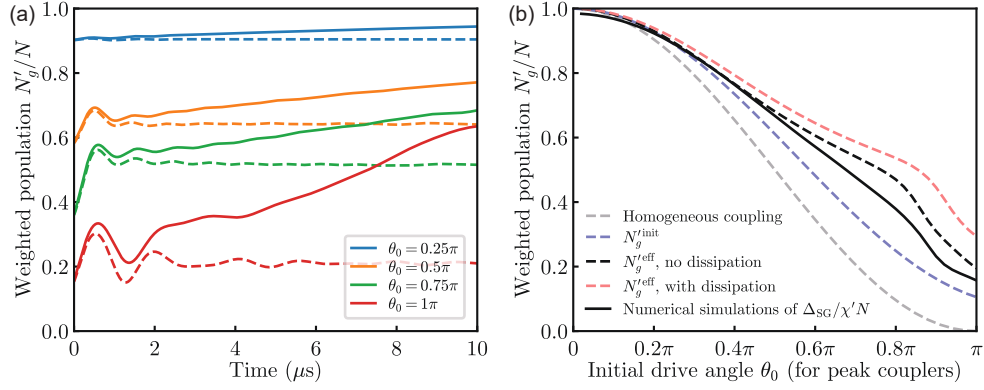


FIG. S2. Analysis of the weighted ground state population N'_g . (a) Time dynamics after an initial Rabi pulse with drive angle θ_0 for peak couplers. Dashed lines are derived from simulations neglecting single-particle and collective dissipation, and solid lines represent a full numerical simulation of the experiment (including dissipation). (b) Comparison of various N'_g measurements vs. θ_0 . The gray dashed trace represents homogeneous coupling, with all other traces assuming uniformly inhomogeneous coupling. The blue dashed trace shows the expected initial-time population according to Eq. (S42). The black and red traces represent N'_g averaged over the time interval $t = 0.5 \mu\text{s}$ to $t = 5 \mu\text{s}$ (the interval used to measure Δ_{SG}), using simulations neglecting and including dissipation, respectively. Finally, the black solid trace is Δ_{SG} computed from numerical simulations with the same method used with the experimental data, normalized to 1 at $\theta_0 = 0$.

In the experiment, we estimate Δ_{SG} by analyzing the oscillation frequency of $\hat{J}^-$ within the measurement interval $t \in [0.5 \mu\text{s}, 5 \mu\text{s}]$. The black dashed line represents the average weighted ground state population in this time interval, which we call the effective ground state population N'_g/eff , using numerical simulations without dissipation. By visual inspection of Fig. S2a, this estimate should be fairly close to the long-time steady-state value for N'_g . We also plot N'_g/eff using simulations including dissipation (red dashed line). Finally, we compare all of these estimates against Δ_{SG} calculated from numerical simulations and normalized to the same units.

One might be tempted to claim that Δ_{SG} should increase in the presence of dissipation since N'_g increases over time. While this is true for collective dissipation, it is not necessarily true when single-particle spontaneous emission is the dominant loss process (as is the case in our experiment). This is because the relationship $\Delta_{\text{SG}} = \chi N_g$ only holds for maximally symmetric states. In general, states with lower angular momentum will also have a smaller Δ_{SG} . This can be seen for homogeneously coupled systems by considering a wavefunction of the form $|\Psi_i\rangle = |J, J^z\rangle_{gb} \otimes |\psi\rangle_d$, which is an eigenstate of the interaction term $\chi \hat{J}^+ \hat{J}^-$ with eigenvalue $E_i = \chi(J(J+1) - J^z(J^z-1))$. Transferring one atom symmetrically from $|b\rangle$ to $|d\rangle$ transforms the wavefunction to $|\Psi_f\rangle = |J - \frac{1}{2}, J^z - \frac{1}{2}\rangle_{gb} \otimes |\psi'\rangle_d$, which can be seen using the Schwinger boson formalism introduced in Sec. II. This is also an eigenstate of the interaction term with eigenvalue E_f , giving a total energy cost equal to

$$\begin{aligned} \Delta_{\text{SG}} = E_i - E_f &= \chi(J(J+1) - J^z(J^z-1)) - \chi\left(\left(J - \frac{1}{2}\right)\left(J + \frac{1}{2}\right) - \left(J^z - \frac{1}{2}\right)\left(J^z - \frac{3}{2}\right)\right) \\ &= \chi(J - J^z + 1) \leq \chi(N_g + 1), \end{aligned} \quad (\text{S43})$$

with equality holding iff $J = \frac{N_b + N_g}{2}$. We conclude that as atoms dephase with fixed J^z , Δ_{SG} decreases.

The takeaway from this analysis is that the red dashed curve in Fig. S2b overestimates Δ_{SG} because it does not take dephasing into account. This is verified by the fact that the computed value of Δ_{SG} (black solid curve) lies below this curve. In the main text, we compare Δ_{SG} against the dissipation-free version of N'_g/eff (black dashed curve) for this reason. In fact, we see that this curve also overestimates Δ_{SG} . This is likely because the initial drive pulse does not place the system in a maximally symmetric state (atoms with different cavity couplings are initialized to a different position on the Bloch sphere).

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